SQL Code

```sql
-- Student Academic Management System Using SQL
use student;
-- Display all students.

SELECT * FROM students;

-- Show all instructors.

SELECT * FROM instructors;

-- Show all courses.

SELECT * FROM courses;

-- Show students from Mumbai.

SELECT student_name
FROM students
WHERE city = 'Mumbai';

-- Display students from the Computer Science department.

SELECT student_name
FROM students
WHERE department = 'Computer Science';

-- Show instructors with more than 10 years of experience.

SELECT instructor_name
FROM instructors
WHERE experience_years > 10;

-- Show courses with 4 credits.

SELECT course_name
FROM courses
WHERE credits = 4;

-- Display students older than 21.

SELECT student_name
FROM students
WHERE age > 21;
--  Aggregate Function Queries

-- Count total students.

SELECT COUNT(*)
FROM students;

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```sql
-- Average student age.

SELECT AVG(age)
FROM students;

-- Maximum instructor salary.

SELECT MAX(salary)
FROM instructors;

-- Minimum instructor salary.

SELECT MIN(salary)
FROM instructors;

-- Total instructor salary.

SELECT SUM(salary)
FROM instructors;
--  GROUP BY Queries

-- Number of students in each department.

SELECT department, COUNT(*)
FROM students
GROUP BY department;

-- Average score per course.

SELECT course_id, AVG(score)
FROM exams
GROUP BY course_id;

-- Total students enrolled in each course.

SELECT course_id, COUNT(student_id)
FROM enrollments
GROUP BY course_id;

-- Average instructor experience by department.

SELECT department, AVG(experience_years)
FROM instructors
GROUP BY department;
--  JOIN Queries

-- Show students and their courses.

SELECT s.student_name, c.course_name
FROM students s
JOIN enrollments e ON s.student_id = e.student_id
JOIN courses c ON e.course_id = c.course_id;

-- Show courses with instructor names.
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SELECT c.course_name, i.instructor_name
FROM courses c
JOIN instructors i
ON c.instructor_id = i.instructor_id;

-- Show students with exam scores.

SELECT s.student_name, c.course_name, e.score
FROM exams e
JOIN students s ON e.student_id = s.student_id
JOIN courses c ON e.course_id = c.course_id;

-- Show students with grades.

SELECT s.student_name, e.grade
FROM students s
JOIN exams e
ON s.student_id = e.student_id;
-- Intermediate Queries

-- Find students who scored above 90.

SELECT student_id
FROM exams
WHERE score > 90;

-- Find courses taught by instructors from Computer Science department.

SELECT course_name
FROM courses
WHERE department = 'Computer Science';

-- List students enrolled in Fall 2023 semester.

SELECT student_id
FROM enrollments
WHERE semester = 'Fall'
AND year = 2023;

-- Find students who enrolled in more than one course.

SELECT student_id, COUNT(course_id)
FROM enrollments
GROUP BY student_id
HAVING COUNT(course_id) > 1;
--  Advanced Queries

-- Find the student with the highest score.

SELECT student_id, MAX(score)
FROM exams;

-- Find the course with highest average score.
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```sql

SELECT course_id, AVG(score)
FROM exams
GROUP BY course_id
ORDER BY AVG(score) DESC;

Find instructors teaching multiple courses.

SELECT instructor_id, COUNT(course_id)
FROM courses
GROUP BY instructor_id
HAVING COUNT(course_id) > 1;

-- Find departments with more than two students.

SELECT department, COUNT(*)
FROM students
GROUP BY department
HAVING COUNT(*) > 2;

-- Find students who received grade 'A'.

SELECT student_id
FROM exams
WHERE grade = 'A';
```